

## Examiner reports

### Q1.

In part (a), the graph was generally well drawn with many students scoring full marks. Inaccuracy with the second section was the main cause of difficulty.

In part (b), most students seemed familiar with the concept “distance = area under the graph”. However, what “under the graph” meant was not well applied. Having obtained  $\frac{1}{2} \times 3 \times 4 = 6$  for the first area, many calculated  $\frac{1}{2} \times 5 \times 6 = 15$  for the second area. Others stated that the vehicle was “back at  $P$ ” after 6.5 seconds, clearly confusing velocity and displacement. Many obtained the correct areas, but added them all together, giving the total distance travelled rather than the distance from  $P$ . A significant minority handled the plus and minus areas well for a straightforward 3 marks.

### Q2.

Surprisingly, only just over half of students got part (a) correct. The most frequent errors seen were numerical slips, writing  $-4$  when the question asked for magnitude and picking the wrong section of the graph. Most seemed to know they had to find a gradient.

Just under half of students scored full marks on part (b). Of the students who knew they had to compare areas above and below the time axis, several used correct but elaborate methods often setting up equations involving time. The most successful students simply counted squares often showing any working on the diagram. Common errors included working with gradients instead of areas or counting incorrect sections of the graph as positive areas when they should have been negative.

### Q6.

This question was also done very well with a high success rate. The main issues that prevented students gaining full marks were arithmetical errors or reading incorrect values from the graph. A few students omitted one section when finding the area. Occasionally, students would incorrectly take the mean of some velocities, for example calculations like  $\frac{5+6+7}{3} = 6$ , but this could not be described as a common approach.

### Q7.

There were very many good solutions to this question, with many candidates gaining full marks. There were some arithmetic errors that were introduced in parts (a) and (b) when calculating the distances. The questions about the displacement and the average velocity in parts (d) and (e) did cause some confusion, particularly for some candidates who tried to use the formula  $s = \frac{1}{2} (u + v) t$ .

### Q8.

This question was also done very well by the candidates. The main issues were in parts (a) and (b). In part (a), a number of candidates made errors when calculating the area under the graph. Often they would have one small error, which meant that their total was incorrect. In these cases, the candidates’ distances were followed through into part (b). Both parts (b) and (c) were done well with very few candidates losing marks. In part (d),

Newton's second law was applied in all but a few cases, but the issue was with the units of the mass. Some candidates converted to kg incorrectly, while others simply found the force as if the mass were 200 kg.

**Q9.**

(a) was usually done correctly. Part (b) saw more errors from candidates, with some failing to subtract the 400 from the 2000 and using 2000 metres in their calculations. A few also simply divided their distance by 18 rather than using a constant acceleration equation. In part (c), many candidates drew a graph with the correct shape, but many did not label the required speeds and times clearly on the axes. In many cases, a scale (for example 4, 8, 12, 16,...) had been put on the axis, rather than just indicating the initial and final speeds. Part (d) was generally done very well, but a few candidates did take the mean of 18 and 32 rather than using the total distance divided by the total time.

**Q10.**

This question was done well by the vast majority of candidates, with many scoring full marks. Although not made often, the most common error was failing to multiply by 0.5, or divide by 2, when finding the distances in parts (b) and (c). Most candidates used a correct approach to find the average speed in part (d). Another error, seen in part (e), was to find 2400 but not subtract the answer to part (c).

**Q11.**

Overall this question was done well, but there were a few issues that appeared with a fair number of scripts. In part (a) some candidates often gave only two times and some gave the times when the speed was a minimum or maximum. Part (b) provided examples of candidates showing insufficient working. For example, some candidates simply wrote  $5 \times 15 = 75$ , which did not provide enough justification to gain any marks. Parts (c) and (d) were generally done very well with the occasional arithmetic error.

**Q12.**

This question was done very well by the vast majority of the candidates. Almost all gained full marks on part (a), but a few did make arithmetic errors. Again, in part (b), almost all

the candidates found the acceleration correctly, although a few did give the answer  $\frac{4}{3}$ . Part (c) did cause a little more difficulty, although there were many correct responses. The two most common errors were either to assume that the tension was equal to the weight or to make a sign error and obtain an equation such as  $400g - T = 400a$ .

**Q13.**

The first two parts of this question were done very well by the vast majority of candidates, but part (c) caused more difficulties. In part (a), the candidates almost always drew a graph with the correct shape, but made errors on the axes. The two most common errors were to use 21 instead of 12 on the time axis and to use an incorrect value on the velocity axis. In part (b), a very small number of candidates made arithmetic errors and a few also used incorrect values.

Part (c) caused more difficulties for the candidates. There were three main incorrect

approaches. The first of these was to assume that the forces on the lift were in equilibrium and hence simply state that the tension was equal to the weight, giving their answer as 2940 N. Some candidates used an incorrect value for the acceleration, which they then included in a three term equation of motion which would have otherwise been correct. The most common approach was to take the acceleration as  $2 \text{ m s}^{-2}$ .

A few candidates used a three term equation of motion with the correct terms but an incorrect sign. This approach would produce a result such as  $T = 2940 - 150 = 2790 \text{ N}$ .

**Q14.**

Again there were frequent sign errors in part (a)(i). However, the rest of the question was popular and mostly done well with many fully complete solutions.

**Q15.**

The sketch graph in part (a) proved a good source of marks although some graphs were drawn without the use of a ruler and intended straight lines were 'wobbly'. Part (a)(ii) was usually successful although some omitted some of the areas in trying to find the total distance. Parts (a)(iii) and (a)(iv) were popular and successful. Part (b)(i) was done well, but part (b)(ii) mostly lacked the required link between the curves of the graph and the more realistic changes in motion.

**Q16.**

A straightforward question, but not always yielding the marks it should. In part (a) many were careless with working, often confusing values of  $u$  and  $v$ , and altering or ignoring incorrect signs in subsequent working. Candidates should be aware that 'show that' requests require convincing correct working leading to the printed result. In part (b) sketches were usually good but sometimes marred by incomplete labelling. Part (c) was done well. Responses in part (d) sometimes referred to irrelevant factors rather than the model of the motion given in the question.

**Q17.**

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**Q18.**

Generally this question was done very well. An error that was observed several times in part (a) was to find the area above the graph, rather than the area below the graph. Part (b) was done very well. Part (c) was also done well, but some candidates rounded their final answer to two significant figures.

**Q19.**

This was done well by most, although there was occasional confusion seen in using the

appropriate distance for parts (b) and (c).

**Q20.**

This question proved both popular and successful for candidates. Part (a) was very well done and part (b) only slightly less so with occasional accuracy errors or misunderstanding regarding the average speed.

**Q21.**

Parts (a)(i) and (ii) were often well done, although there was some use of the 'straight string' method, and some lack of understanding of relevant forces acting on the particle on the horizontal surface.

Part (a)(iii) proved discriminating as expected, but part (b) revealed a lack of understanding of the two stages of the motion.

**Q22.**

The majority of candidates were successful in part (a) but too many could not attempt part (b) with an appropriate method. Part (c) was sometimes omitted, and some criticisms did not relate to the model given.

**Q23.**

This question was very well answered.

**Q24.**

Part (a) was mostly completed successfully, showing a pleasing improvement in algebraic skills. Part (b) proved more discriminating. Many candidates found sketch graphs helpful

in understanding the problem. The most direct method was the use of  $s = \frac{1}{2}(u+v)t$ , but many were successful with longer methods involving the retardation.

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**Q25.**

Parts (a) and (b) of this question proved very successful. Some candidates had difficulty in appreciating the nature of the motion after the string broke and chose an incorrect acceleration in part (c). Graphs in part (d) were generally good.

**Q26.**

Diagrams in part (a) were good, but algebraic expressions produced in part (b) were poor. The use of the simple principle 'speed = distance / time' proved beyond many candidates. Many seemed unaware of what constitutes an 'equation', omitting the equals sign. There were, however, some excellent solutions to this question, showing clear and concise work.

**Q27.**

The majority of candidates made a good attempt at this question and there were many completely correct solutions. In part (a), some of the candidates considered the three

stages of the motion, but made errors when calculating one or more of these distances. In part (b), a very small number of candidates confused the numbers and gave the reciprocal of the acceleration as their answer. The majority of force diagrams in part (c) were correct, but the most common error was to omit one force.

Calculating the force in part (c) caused the most difficulties. Some candidates made a sign error in their equations, but the most common error was to simply multiply the acceleration by the mass, without taking account of the resistance force.



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